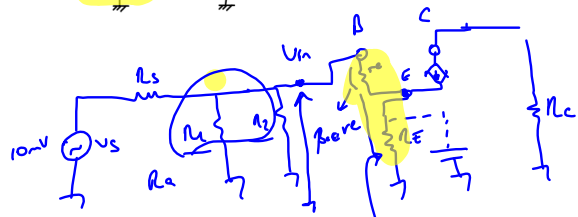
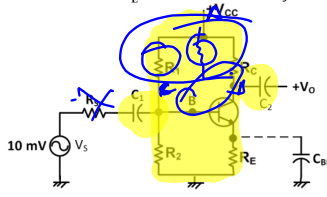


kazancını bulunuz. $I_E = 2.4 \text{ mA}$ olarak verilmiştir.

$\beta_{ac} = 140$, $R_S = 300 \Omega$, $R_1 = 22 \text{ k}\Omega$, $R_2 = 4.7 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, $R_E = 600 \Omega$

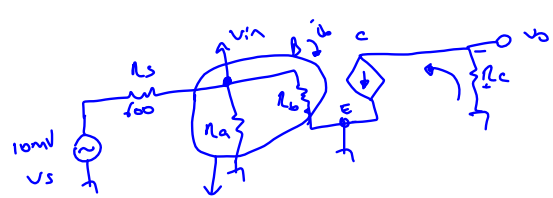
$$r_e = \frac{V_T}{I_E} = \frac{26 \text{ mV}}{2.4 \text{ mA}} =$$



de kondansatör yoldan

$$R_a = \frac{R_1 \cdot R_2}{R_1 + R_2} = 3.82 \text{ k}$$

$$R_b = \beta \cdot (r_e + R_E) = 85.5 \text{ k}$$



$$R_{in} = \frac{R_a \cdot R_b}{R_a + R_b} = 3.702 \text{ k}$$

$$V_{in} = \frac{V_S}{R_S + R_{in}} \cdot R_{in} = 9.2 \text{ mV}$$

$$i_b = \frac{V_{in}}{R_b} = 0.108 \text{ mA}$$

$$V_o = i_c \cdot R_C = -\beta_{ac} \cdot i_b \cdot R_C = -15.145 \text{ mV}$$

$$A_v = \frac{V_o}{V_{in}} = -\frac{15.145 \text{ mV}}{9.2 \text{ mV}} = -1.63$$

Kondansatör varken R_E by-pass olur,

$$R_b = \beta_{ac} \cdot r_e$$

$$R_{in} = \frac{R_a \cdot R_b}{R_a + R_b} = 1.089 \text{ k}$$

$$V_{in} = \frac{V_S}{R_S + R_{in}} \cdot R_{in} = 7.8 \text{ mV}$$

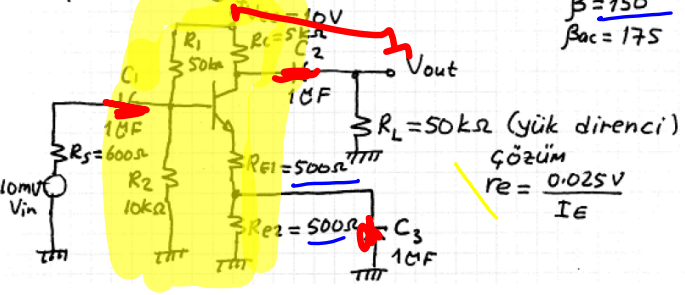
$$i_b = \frac{V_{in}}{R_b} = 5.17 \text{ mA}$$

$$V_o = -723.83 \text{ mV}$$

$$A_v = \frac{V_o}{V_{in}} = -92.308$$

ÖRNEK 2: Aşağıda verilen devrede gerilim kazancını ve

toplam çıkış gerilimini bulunuz



$\beta = 150$
 $\beta_{ac} = 175$

$\beta \cdot (R_{E1} + R_{E2}) \geq 10 \cdot R_L$

$150 \cdot (1k) \geq 10 \cdot 10k$
 $150k \geq 100k$
✓

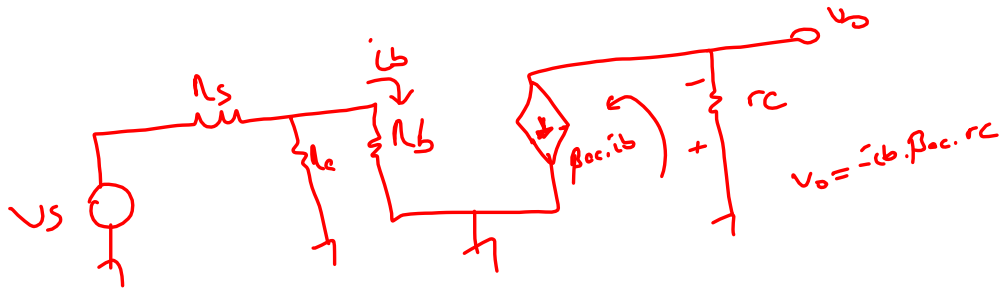
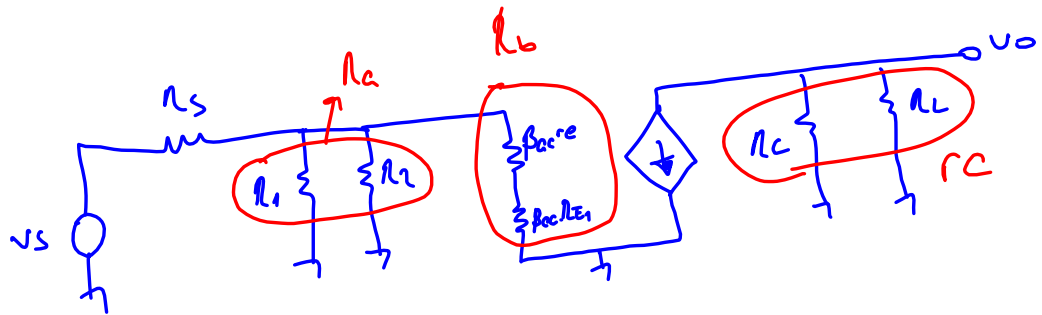
$V_B = \frac{V_{CC}}{R_1 + R_2} \cdot R_2 = 1,66V$

$V_E = V_B - 0,7 = 0,96V$
(V_{BE})

$I_E = 0,96 \text{ mA}$

$V_{CE} = V_{CC} - I_E \cdot (R_C + R_E) = 4,2V$ ✓

$r_e = \frac{V_T}{I_E} = \frac{25 \text{ mV}}{0,96} = 25,81 \Omega$
 $V_{BE} < V_{CE} < V_{CC}$ ✓

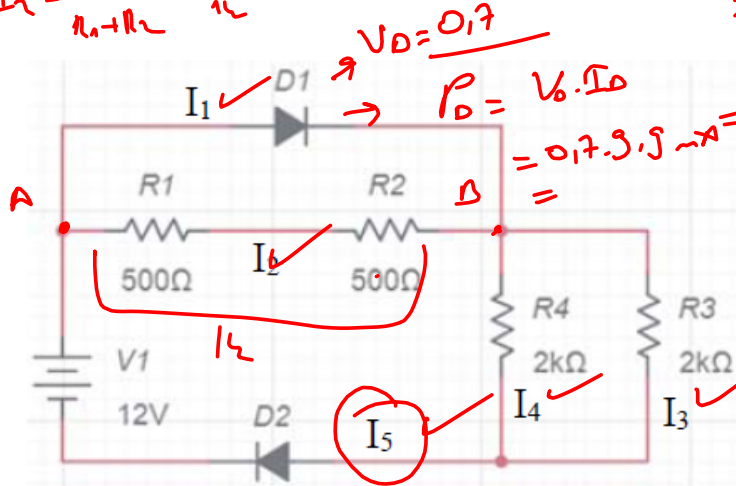


$V_{in} = i_b R_b$
 $V_{in} = \frac{V_s}{R_s + (R_1 || R_2 || R_3 || R_b)} \cdot (R_1 || R_2 || R_3 || R_b) \Rightarrow V_{in} = \frac{V_s}{R_s + R_{in}} \cdot R_{in}$
 $V_{in} = 9,2 \text{ mV}$

$i_b = \frac{V_{in}}{R_b} \approx 0,1 \text{ mA}$

$$I_2 = \frac{V_D}{R_1 + R_2} = \frac{0,7}{1k} = 0,7 \text{ mA}$$

$$I_2 = \frac{V_D}{R_1 + R_2} = \frac{0,7V}{1k} = 0,7 \text{ mA}$$



$$V_D = 0,7$$

$$P_D = V_D \cdot I_D$$

$$= 0,7 \cdot 9,3 \text{ mA} = 6,51 \text{ mW}$$

$$I_1 + I_2 = I_3 + I_4$$

$$I_1 + 0,7 = 10,6$$

$$I_1 = 9,9 \text{ mA}$$

$$I_3 + I_4 = I_5$$

$$I_3 = I_4 \Rightarrow I_3 = 5,3 \text{ mA}$$

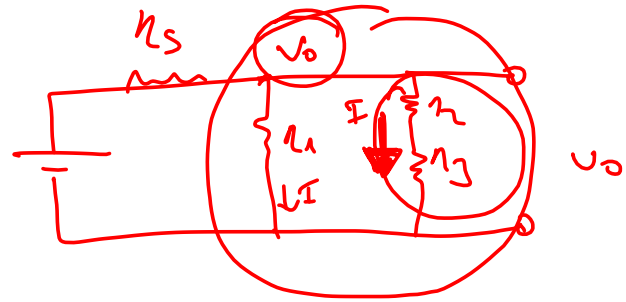
$$I_4 = 5,3 \text{ mA}$$

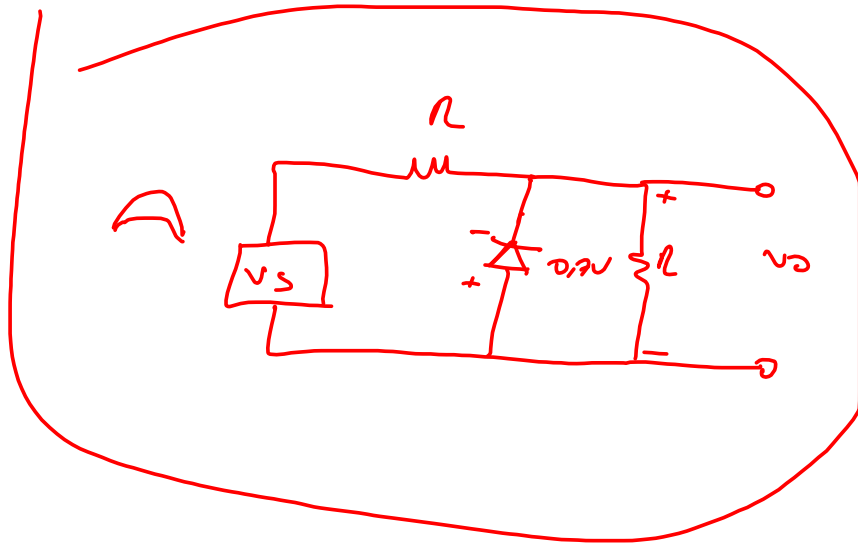
$$V_D = I_2 \cdot (R_1 + R_2)$$

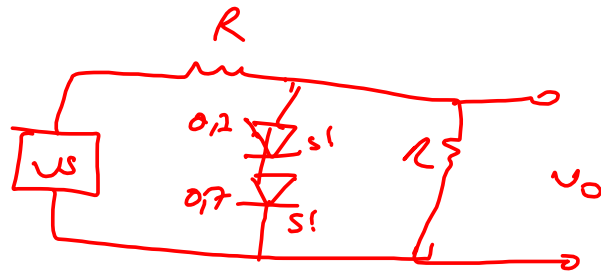
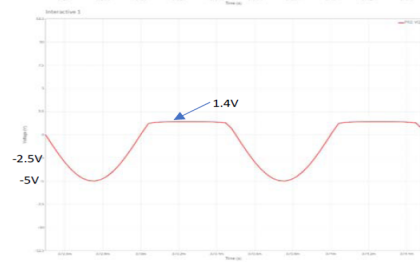
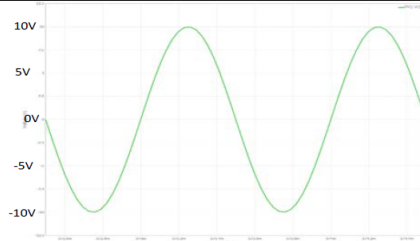
$$V_1 - V_D - I_5 \cdot (R_3 // R_4) - V_D = 0$$

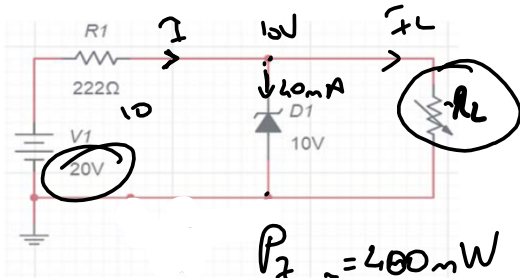
$$12 - 0,7 - I_5 \cdot 1k - 0,7 = 0$$

$$I_5 = \frac{10,6}{1k} = 10,6 \text{ mA}$$









$$P_{Z_{max}} = 400 \text{ mW}$$

↓

$$I_{Z_{max}} = \frac{400}{10 \text{ V}} = 40 \text{ mA}$$

$$I = I_Z + I_L$$

$$\downarrow$$

$$45,045 \text{ mA} = 40 \text{ mA} + I_L$$

$$I_L = 5,045 \text{ mA}$$

$$I_L \cdot R_L = 10 \text{ V}$$

$$R_L = \frac{10 \text{ V}}{5,045 \text{ mA}} = 1,984$$

for Zener load calculation

$$V_{R_L} = \frac{V_1}{R_1 + R_L} \cdot R_L$$

↓

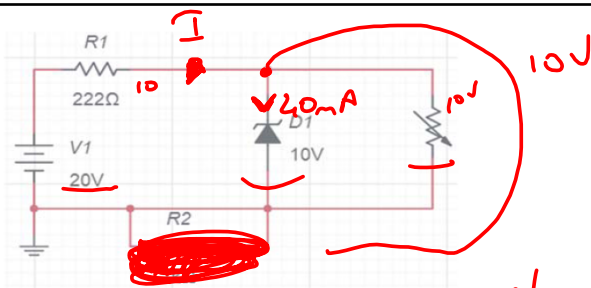
$$10 = \frac{20 \text{ V}}{222 \Omega + R_L} \cdot R_L$$

$$2220 + 10 R_L = 20 R_L$$

$$10 R_L = 2220 \Omega$$

$$R_L = 222 \Omega$$

$$I_{L_{min}} = \frac{V_{in}}{R_L + R_1} = \frac{20 \text{ V}}{444} = 45 \text{ mA}$$



zener positif

$$R_L = 222 \Omega \quad (10V \text{ olmas\u0131 i\u00e7in})$$

$$I_L = \frac{V_1}{R_L} = 45,045 \text{ mA}$$

$$P_{Z_{max}} = 400 \text{ mW}$$

$$I_{Z_{max}} = \frac{P_{Z_{max}}}{V_Z} = \frac{400 \text{ mW}}{10V} = 40 \text{ mA}$$

$$I = I_Z + I_L$$

$$45,045 \text{ mA} = 40 \text{ mA} \rightarrow$$

$$I_L = 5,045 \text{ mA}$$

$$R_{L_{max}} = \frac{V_Z}{I_L} = \frac{10V}{5,045} = 1,98 \Omega$$